

Notes

Chapter 2 – Classical Encryption Techniques

1. Symmetric Cipher Model

Main idea: Same secret key is used for both encryption and decryption.

Definitions:

- Plaintext = Original message
- Ciphertext = Encrypted message
- Key = Secret value controlling encryption
- Encryption and decryption algorithms are public.

Cryptanalysis: Recover plaintext or key without knowing the secret key.

Brute Force Attack: Try every possible key until one works.

2. Substitution Ciphers

Main idea: Replace letters with other letters.

Order of letters stays the same.

Example:

$$HELLO \rightarrow KHOOR$$

Caesar Cipher

Shift every letter by a fixed amount.

Example (Shift = 3):

$$A \rightarrow D, \quad B \rightarrow E, \quad C \rightarrow F$$

Encryption formula:

$$C = (P + k) \bmod 26$$

where

- P = plaintext
- k = key
- C = ciphertext

Weakness: Only 25 useful keys $\Rightarrow$ brute force is easy.

Monoalphabetic Cipher

Uses one random substitution alphabet.

Example:

ABCDEFGHIJKLMNOPQRSTUVWXYZ

QWERTYUIOPASDFGHJKLZXCVBNM

More secure than Caesar but still vulnerable due to Frequency Analysis
Most common English letters:

E > T > A > O > I

Attacker compares ciphertext frequencies with English frequencies.

Playfair Cipher

- Encrypts 2 letters together.
- Uses a 5×5 key matrix.
- More secure than single-letter substitution.

Hill Cipher

Uses matrix multiplication.

Steps:

1. Convert letters to numbers.
2. Multiply using key matrix.
3. Take modulo 26.
4. Convert back to letters.

Introduces linear algebra into cryptography.

Vigenère Cipher

Uses multiple Caesar ciphers.

Example key:

KEYKEYKEY

Each letter is shifted differently.

Much stronger than monoalphabetic substitution.

3. Transposition Cipher

Letters remain the same.
Only their positions change.
Example:

$$HELLO \rightarrow LHOEL$$

Difference:

- Substitution $\Rightarrow$ letters change.
- Transposition $\Rightarrow$ positions change.

4. Product Cipher

Modern encryption combines

- Substitution
- Transposition

repeated many times.
DES and AES are product ciphers.

5. Rotor Machines

Mechanical encryption machines.
Rotor rotates after every key press.
Same plaintext letter can encrypt differently each time.
Example:

$$A \rightarrow Q, \quad A \rightarrow M, \quad A \rightarrow T$$

Harder to attack than Caesar cipher.

6. Steganography

Different from encryption. It's main purpose is to hide the existence of the message.
Examples:

- Images
- Audio
- Video
- Invisible ink

Encryption hides the content.
Steganography hides the message itself.